Temma Driver

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Introduction

The CCDASTRO Takahashi Temma ASCOM driver lets ASCOM-compatible astronomy applications control Takahashi Temma mounts through the mount's serial interface. Typical client applications include N.I.N.A., TheSky, ACP, and other programs that use the ASCOM Telescope interface.

The driver supports connection, coordinate reporting, slewing, aborting a slew, synchronizing coordinates, sidereal/lunar/solar/King tracking rates, manual motion, pulse guiding, parking, and unparking.

Revision History

Version 1.0.20

- Preserves and reports the saved ASCOM park position after reconnecting.
- Keeps Temma coordinate polling from overwriting the reported park position while the driver is parked.
- Refreshes the driver's coordinate cache after a successful Sync operation.

Version 1.0.19

- Corrected counterweight-down and counterweight-west startup references.
- Uses the configured site latitude and local sidereal time to establish the correct celestial coordinate for those physical mount positions.

Version 1.0.18

- Improved slew-completion detection by requiring observed mount motion to settle before reporting `Slewing = false`.

Version 1.0.17

- Restored reliable operation after Abort Slew by using the Temma stop command
  without an unsupported post-abort status query.

Getting Started

System requirements

Software

1.A computer running in Windows 10/11 (64-bit).

2.ASCOM Platform 6 (or later) must be installed.

3..Target Net Framework 4.7.2.

Hardware

1. Takahashi Temma 2, or Temma 2M mount.

2. A computer with a serial port or USB/Ethernet to serial port converter.

Note: Users should avoid Keyspan USA-19HS serial adapters, BSOD crashes occur when used with the ASCOM Temma driver. The crash dump file points to the Keyspan driver, this issue was reported to Tripp Lite on 12/12/10. BSOD occur from flaws in kernel level drivers, the ASCOM TEMMA driver is user level, indicating a hardware fault or a buggy kernel mode Keyspan driver.

3. A Serial cable to connect a PC to the telescope.

4. Power Source

Select the appropriate power setting from the 'Power' drop-down box. You must have the appropriate 24 volt power supply to run at the 24 volt high slew rates.

High Speed Drive	Working Voltage	
	DC12V	DC24V
EM-400 Temma2	250X	500X
EM-10 Temma2 Jr.	120X	250X
EM-200 Temma2 Jr.	120X	250X

EM-200 Temma2	350X	700X
NJP Temma2	175X	350X
EM-500 Temma2	260X	520X
EM-200M	700X	Not Supported
EM-400M	500X	Not Supported
EM-12M	150X	Not Supported

Installing

Install the driver using its release installer, download here.

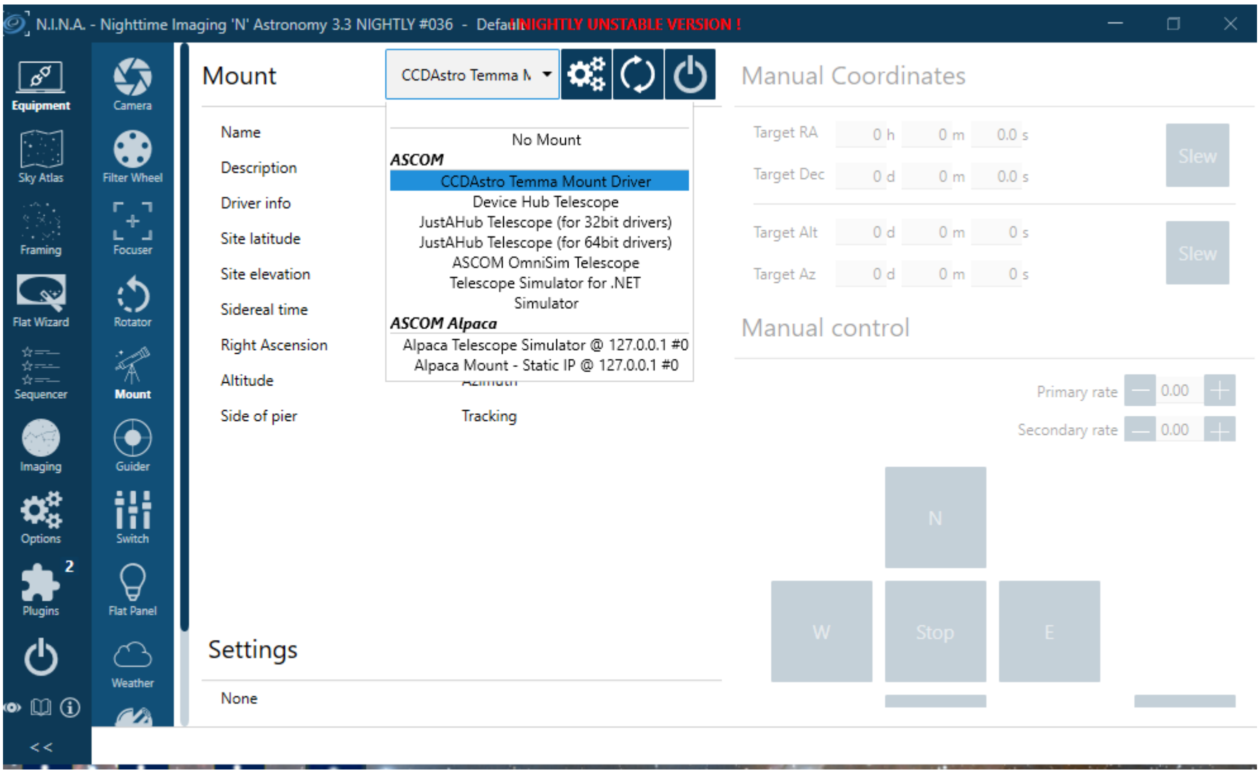
Using the Temma Driver with Client Apps

To use the driver select it in an ASCOM application.

For example, In N.I.N.A., TheSky, or another ASCOM client:

1. Select 'CCDASTRO Temma Mount Driver' as the telescope driver.
2. Open Setup and select the COM port, site information, mount configuration, startup orientation, and optional park settings.
3. Physically place the mount in the selected startup orientation.
4. Connect the telescope.
5. Confirm the reported RA/Dec and Alt/Az before issuing a slew.
6. Use the client’s normal Slew, Abort, Tracking, Park, and Unpark controls.

The client program handles planning and safety decisions such as meridian avoidance, dome coordination, and weather monitoring. Configure those client features separately.



Getting help and reporting driver issues

In order of preference potential driver problems can be reported via several methods:

E-mail to ccdastroinc@gmail.com.

IMPORTANT: When reporting a potential ASCOM driver bug here are some general guidelines:

1. The first aim of a bug report is to let us see the failure with our own eyes. Please give us detailed instructions so that we can make it fail for ourselves.

2. The second aim of a bug report is to describe what went wrong. Describe everything in detail. State what you saw, and also state what you expected to see. Write down the error messages, especially if they have numbers in them. Better yet, take a screenshot!

3. By all means try to diagnose the fault yourself if you think you can, but if you do, you should still

report the symptoms as well.

4. Be ready to provide extra information if we need it, like trace logs.

5. If other programs are involved state the EXACT version number of each program. Usually you can find this information in the programs Help->About menu item.

6. Write clearly. Say what you mean, and make sure it can't be misinterpreted!

Thank you in advance for following these guidelines! For more details please see this link:

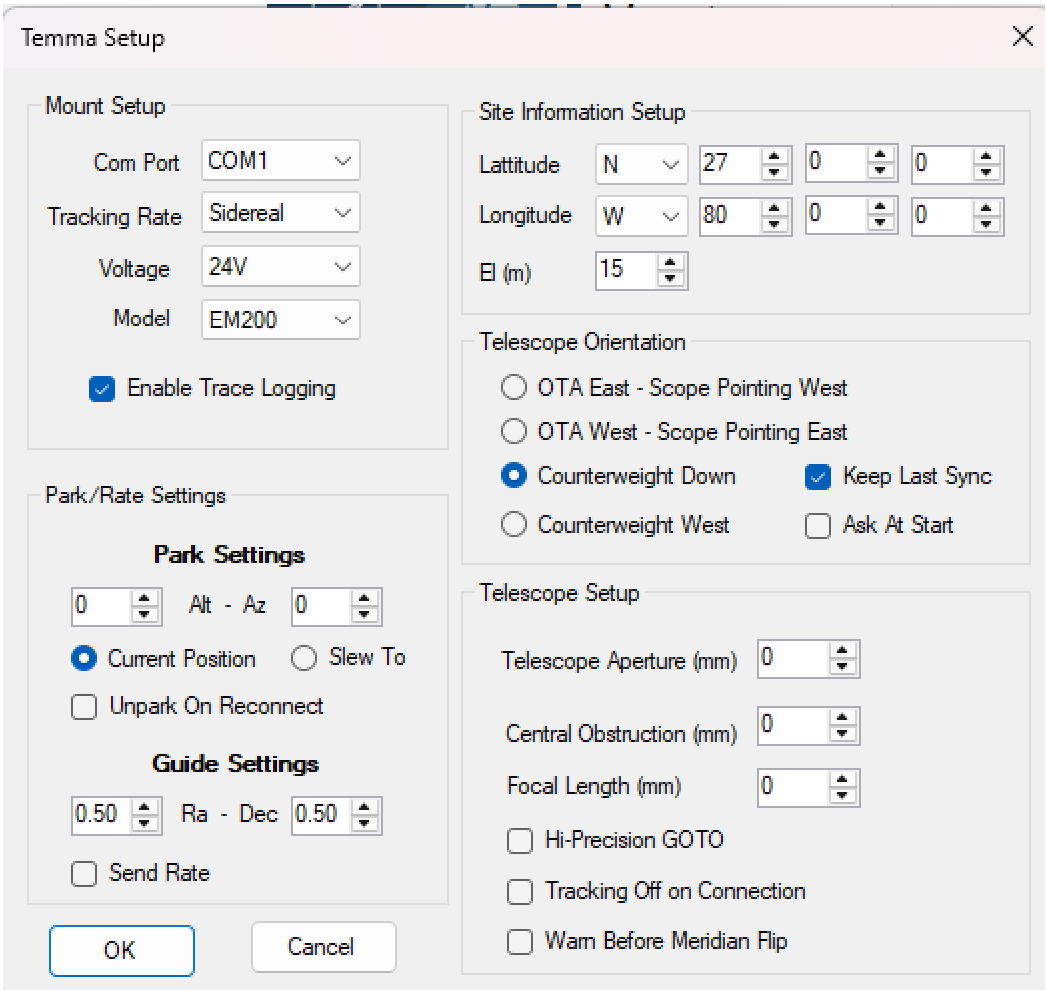
Github

Temma Mount Setup and Features

ASCOM Setup

Telescope Setup window:

Detailed explanations of each field follows in the next sections.



Serial Port

Select the Windows COM port for the mount or simulator. Do not open the same COM port in another program, including the Arduino Serial Monitor, while the ASCOM driver is connected.

The Temma serial settings are fixed by the driver:

Setting	Value
Baud rate	19200
Data bits	8
Parity	Even
Stop bits	1
Line ending	CR/LF
DTR	Enabled

Initialization

Temma Setup and Initialization Procedure

Before connecting, physically place the mount in the orientation selected in the Setup window. The driver uses the selected reference to synchronize its initial coordinate system with the physical mount.

Available startup references are:|

| OTA East - Scope Pointing West | Telescope is on the east side of the pier and points west. |

| OTA West - Scope Pointing East | Telescope is on the west side of the pier and points east. |

| Counterweight Down | Counterweight shaft points down; the right-ascension axis points toward the true celestial pole. |

| Counterweight West | Counterweight shaft is horizontal toward the west. |

Counterweight Down is not a horizon-pointing position. At a northern site, the optical axis is near the north celestial pole, approximately at an altitude equal to the site's latitude and azimuth near true north. Azimuth is geographic/true azimuth, not magnetic compass bearing.

For a reliable connection, do not change the mount's position manually after selecting a reference and before the driver has completed its initial synchronization.

Synchronizing	Procedure
To synchronize with the optical tube assembly on the East side of the mount, Scope pointing West	1. Loosen the clutches on the mount and move the telescope to the East side of the mount. 2. Center a known star that is on the West side of the Meridian in the field of view. 3. Identify the star in your planetarium software. 4. Sync on the star. 5. An on screen dialog will notify you of the correct side of the meridian to select the sync star.
To synchronize with the optical tube assembly on the West side of the mount, Scope pointing East	1. Loosen the clutches on the mount and move the telescope to the West side of the mount. 2. Center a known star that is on the *East* side of the Meridian in the field of view. 3. identify the star in your planetarium software. 4. Sync on the star. 5. An on screen dialog will notify you of the correct side of the meridian to select the sync star.
Scope Pointed at Pole, Counter-weight down position	This method is intended for remote users. The CW down method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. It is recommended that the subsequent sync be accomplished using ACP (Pinpoint) to plate solve a CCD image after

	the initial alignment. Obviously, a manual/visual sync can be used if desired.
Scope Pointed at Pole, Counter-weight West	The CW West method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. Place the CW on the West side of mount with the CW shaft and OTA parallel to the ground.

Site Setup

Site Setup

Enter the observing site latitude, longitude, and elevation accurately. Latitude is positive north of the equator; longitude follows the ASCOM convention used by the setup dialog. Site latitude and longitude affect coordinate conversion, Altitude/Azimuth reporting, and startup reference calculations.

The driver obtains sidereal time from the configured site and the computer's time. Ensure that Windows date, time, time zone, and daylight-saving setting are correct before initializing the mount.

Telescope

Telescope aperture, central obstruction, and focal length are supplied toASCOM clients as descriptive optical information. Enter the actual values for the telescope when they are known.

Note: these fields are not specifically used by the driver but they are passed up to clients if the client asks for them.

Mount Model and Voltage

Choose the Temma mount model and the mount supply voltage that match the controller. These settings are used by the driver when it initializes the mount and selects the appropriate motion behavior.

Trace Logging

Enable **Trace Logging** when diagnosing a connection or command problem. ASCOM trace files record sent and received protocol traffic and are the most useful information to include with a driver issue report. Disable tracing after diagnosis if the extra detail is not needed.

Connecting

When Connect is selected, the driver opens the configured COM port, initializes communication with the Temma controller, verifies coordinate response, and applies the selected startup reference. The driver then reports the current Right Ascension, Declination, Altitude, Azimuth, side of pier, and tracking state to the ASCOM client.

If the mount was previously marked parked and **Unpark On Reconnect** is not selected, it remains parked after reconnect and reports the saved park position. Unpark the mount before requesting a slew or manual movement.

Slewing

An ASCOM client can set Target Right Ascension and Target Declination and call Slew, or can issue a synchronous slew. The driver sends the Temma coordinate and GOTO commands, verifies that motion begins, then reports `Slewing = true`.

The driver does not declare the slew complete simply because the reported coordinate is close to the target. It waits until reported coordinate motion has settled, which avoids prematurely telling the client that tracking has resumed.

If a slew cannot begin, verify that the mount is initialized, not parked, not in standby, and is receiving a valid target within its operating limits.

Abort Slew

Abort Slew sends the Temma stop command and clears the ASCOM slewing state. After an abort, the driver is ready to accept a new slew request. A stopped mount normally resumes the tracking state selected by its Temma controller configuration.

Manual Motion and Guiding

ASCOM MoveAxis operations provide manual right-ascension and declination motion. Stop manual motion through the client before starting another operation.

PulseGuide provides short timed guiding corrections in the requested direction. Configure the RA and Declination guide rates in the Setup window to match the guide performance of the mount and telescope. The **Send Rate** option controls whether those guide-rate settings are sent to the Temma controller when appropriate.

Parking and Unparking

ASCOM park position is configured in the Setup window as altitude and azimuth. The driver can park either by treating the current position as park, or by slewing to the configured park position, according to the selected park method.

The Temma protocol has no dedicated physical Park command. The driver follows ASCOM semantics by stopping any motion, disabling tracking through Temma standby, and retaining the ASCOM parked state and saved park coordinates.

While parked:

- The driver reports the saved park position rather than allowing raw mount coordinate polling to replace it.
- Slews and manual movement should not be requested until Unpark is called.
- The reported tracking state is off.

Unpark clears the driver's parked state and restores tracking by taking Temma out of standby. If **Unpark On Reconnect** is selected, the driver automatically unparks after reconnecting.

At exactly 90 degrees altitude (the zenith), azimuth is mathematically undefined. For a simulator or repeatable test park position, use an altitude such as 85 to 89 degrees if a stable azimuth display is important.

Temma Setup Dialog

Guide Correction Speeds

Guide Rates

Set

Enter the right ascension and declination correction speed (as a percentage of the Sidereal rate, 10 - 99%). This setting is used for the slow speed moves or for guiding.

Power Voltage

Displays the mount's
voltage.

Troubleshooting

Troubleshooting:

The telescope crosshair in my planetarium software jumps	The coordinates reported by the Temma control system (when the telescope is not slewing) can vary by one arcsecond. This causes the telescope cross hairs to "jump" between the reported positions. The Temma protocol reports coordinates to the nearest second in right ascension (15 arcseconds at the 0 degrees declination) and to the nearest 6 arcseconds in declination. The Temma's control system accepts declination coordinates to the nearest 12 arcseconds, so, at small fields of view, the cross hairs will not exactly match the coordinate that was used to synchronize the telescope These are limitations of the Temma protocol.
The telescope crosshair in TheSkyX does not move when slewing or syncing	Increase the crosshair update frequency to 1000 ms or more.
Connection times out	- Confirm the correct COM port in Windows Device Manager. - Confirm no other application has the port open. - Check the serial cable, adapter, and any null-modem requirement for the actual mount wiring. - Verify 19200 baud, 8 data bits, even parity, and 1 stop bit when testing with a terminal program.

	- Enable trace logging and inspect the ASCOM log for transmitted \`E\` commands and received coordinate responses.
Driver connects but coordinates are unexpected	- Recheck site latitude, longitude, computer time, time zone, and daylight saving time. - Confirm the physical telescope orientation matches the Setup selection. - Remember that Counterweight Down points the RA axis toward the true celestial pole; it is not generally Altitude 90 degrees. - Reinitialize from a known physical reference, then Sync only if the reported coordinate is known to be wrong.
A new slew will not start after Abort	- Update to version 1.0.17 or later. - Wait briefly for the mount to stop before issuing the next client command. - Check trace logging for a stop command followed by the new target and GOTO command.
Client remains "slewing" after the mount has stopped	- Update to version 1.0.18 or later. - Allow the coordinate polling interval to observe settled mount motion. - Enable trace logging and provide the log with the approximate time of the slew if the problem persists.
Parked coordinate display changes	- Update to version 1.0.20 or later. - Check that the driver is actually parked, rather than merely tracking off. - Avoid a park altitude of exactly 90 degrees if stable azimuth is required.

Getting Help and Reporting Issues

When reporting a problem, include:

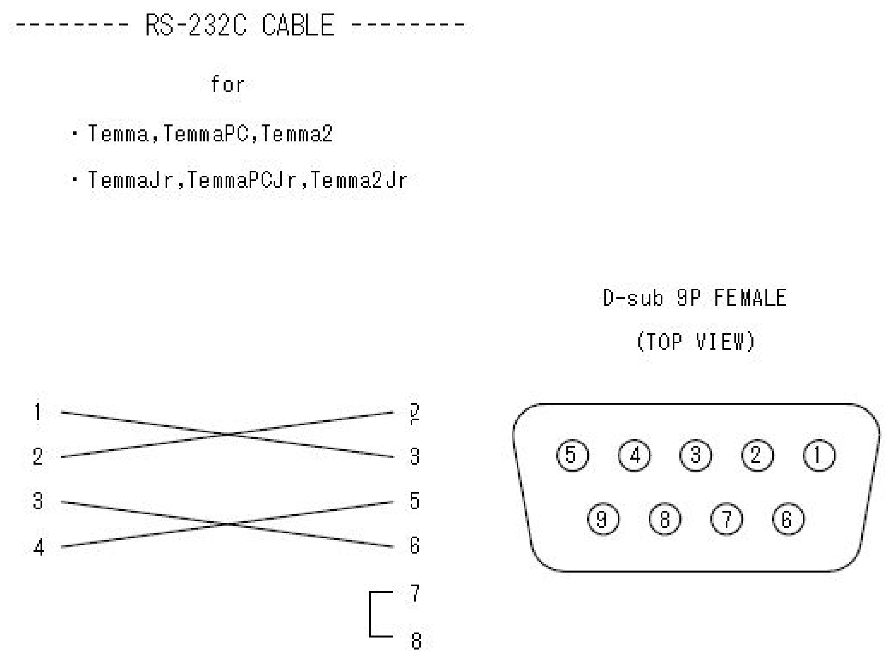
- Driver version.
- ASCOM Platform version and Windows version.
- Mount model, COM port type, and cable/adapter details.
- Site and startup-reference selection, where relevant.
- Client application and its version.
- A clear sequence of actions that reproduces the problem.
- ASCOM trace logs covering the failure.

Email to submit help ticket ccdastroinc@gmail.com

Do not include private observatory coordinates or unrelated personal data in a public report.

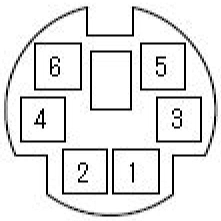
Resources

Cable Drawings



-------- AUTO GUIDE CONNECTOR --------

- USD,USDⅡ,USD3
- Temma,TemmaPC,Temma2
- TemmaJr,TemmaPCJr,Temma2Jr
- PD-7XY,PD-8XY



MINI DIN 6P FEMALE

(TOP VIEW)

No.	
1	NC
2	RA(+)
3	RA(-)
4	DEC(↑)
5	DEC(↓)
6	GND

Driver Updates

Get the most current driver here.